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Fundamental of data Science

Facets of Data

Data has many characteristics or facets that describe its nature, quality and usefulness. understanding these facets helps in collecting, managing, analysis and utilizing data effectively for better insights

1 Structure

Refers to the organization and format of data

- Structured - Organized in rows and columns (e.g, SQL, spreadsheet)
- Semi. structured - Has tags or markers to separate elements (e.g., JSON)

{Key : values}
<tag> ... </tag>

- Unstructured - No Predefined format, Such as free text, images, audio, video, emails, documents.

2 Source

Refers to where the data is generated or collected

- Internal - From within an organization (e.g., sales, employee data, transactions).
- External - From outside Sources (eg., social media, APIs, Market data, government data).
- Primary vs secondary - collected first hand or orbtained for others.
  - Primary : Surveys, sensors, experiments
  - Secondary: Reports, research, databases.

3 Type

Refers to the nature of data values.

- Qualitative (categorical) - Describes qualities or categories (e.g., color, gender, Product type)
- Quantitative (Numerical) - Represent numbers that can be measured or counted

4 Time

Refers to the time aspect of data

- Historical - Data from the Past used for analysis of trends.
- Real-Time - Data generated and streamed in real-time
- Time - series - Data Points collected at different time intervals

5 Quality

Refers to the reliability and accuracy of data

- Accuracy - Correct and error-free data.
- Completeness - No missing values or gaps.
- Consistency - Same format and meaning across datasets.
- Timeliness - up-to data and available when needed.

6 Usage

Refers to how data is used.

- Operational - used in day to day operations (eg., order Processing, inventory management).
- Analytical - used for decision making through analysis, reports and dashboards
- Strategic - used for long-term planning, forecasting and building AI/ML models

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